

RESEARCH ARTICLE

WILEY

An efficient online citrus counting system for large-scale unstructured orchards based on the unmanned aerial vehicle

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Funding information

National Natural Science Foundation of China, Grant/Award Number: 32071912; The Open Competition Program of Top Ten Critical Priorities of Agricultural Science and Technology Innovation for the 14th Five-Year Plan of Guangdong Province, Grant/Award Number: 2022SDZG03; Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation ("Climbing Program" Special Funds), Grant/Award Numbers: pdjh2021a0070, pdjh2022b0079

Abstract

The accurate detection and counting of fruits in natural environments are key steps for the early yield estimation of orchards and the realization of smart orchard production management. However, existing citrus counting algorithms have two primary limitations: the performance of counting algorithms needs to be improved, and their system operation efficiency is low in practical applications. Therefore, in this paper, we propose a novel end-to-end orchard fruit counting pipeline that can be used by multiple unmanned aerial vehicles (UAVs) in parallel to help overcome the above problems. First, to obtain on-board camera images online, an innovative UAV live broadcast platform was developed for the orchard scene. Second, for this challenging specific scene, a detection network named Citrus-YOLO was designed to detect fruits in the video stream in real-time. Then, the DeepSort algorithm was used to assign a specific ID to each citrus fruit in the online UAV scene and track the fruits across video frames. Finally, a nonuniform distributed counter was proposed to correct the fruit count during the tracking process, and this can significantly reduce the counting errors caused by tracking failure. This is the first work to realize online and end-to-end counting in a field orchard environment. The experimental results show that the F1 score and mean absolute percentage error of the method are 89.07% and 12.75%, respectively, indicating that the system can quickly and accurately achieve fruit counting in large-scale unstructured citrus orchards. Although our work is discussed in the context of fruit counting, it can be extended to the detection, tracking and counting of a variety of other objects of interest in UAV application scenarios

KEYWORDS

citrus, counting system, DeepSort, online, UAV, YOLOV5

1 | INTRODUCTION

China is a large producer and consumer of fruit, and its annual fruit planting area and output rank first in the world (He, 2020). In the process of fruit production and processing, preproduction yield estimation is an extremely important task. Accurate orchard yield prediction can not only provide effective information with which to

assist managers' and farmers' decision-making in different stages such as harvest, storage, transportation logistics, and marketing, but it can also assist them in making appropriate management adjustments for the next production cycle to further improve orchard fruit yield and quality. Although yield prediction is important, due to the lack of efficient and accurate fruit yield prediction methods, it is generally carried out by farmers based on past experience, manual

counting and historical data collected by farmers (Rahneemoonfar & Sheppard, 2017). These methods are time-consuming, costly, inefficient and often subjective and destructive. In particular, accuracy is difficult to guarantee using methods based on artificial prediction in orchards with high spatial variability. Therefore, realizing the intelligent yield estimation of orchards is an urgent problem that needs to be solved, and it is of great significance in respect of improving the profit of the fruit industry.

An important prerequisite of intelligent orchard yield estimation is the accurate computation of orchard fruit numbers. Therefore, many scholars have devoted their research to automatic yield mapping processes. With the development of computer vision algorithms, fruit counting for a single image has been realized. To extend the counting method to trees and entire orchards, some scholars (Bargoti & Underwood, 2017; Das et al., 2015; Roy, Dong, et al., 2018; Roy, Kislal, et al., 2018; Wang et al., 2013) collected images or videos from one or both sides of a tree, then used algorithms to detect and count fruits offline. In addition, in recent years, some scholars (Apolo-Apolo, Martínez-Guanter, et al., 2020; Apolo-Apolo, Pérez-Ruiz, et al., 2020; Häni et al., 2020) attempted to combine three-dimensional reconstruction algorithms such as structure from motion (SFM) and simultaneous localization and mapping (SLAM) to undertake the fruit counting task. Although the above methods have good fruit counting accuracy, they usually require expensive equipment, and a step-by-step and complex processing flow. In particular, it can take more than 10 h to process three-dimensional information during the specific steps of such methods, which leads to an inability to perform subsequent counting operations coherently. That is, each step of the yield estimation algorithms is separated, thus affecting the operational efficiency of orchard counting/yield estimation. Moreover, such methods cannot meet the requirements of efficient and real-time operations in smart agriculture in the future.

In existing fruit counting research, image acquisition has often been realized with cameras held by people or attached to cars. This means that the labor is still inefficient, because the photography requires a specific platform, resulting in a time-consuming method and possible aggravation of soil compaction problems (Apolo-Apolo, Martínez-Guanter, et al., 2020). In contrast, using an unmanned aerial vehicle (UAV), with its high flexibility and low cost, is a good solution. At present, UAVs have been successfully applied to precision agriculture for monitoring and counting animals and fruits (Apolo-Apolo, Martínez-Guanter, et al., 2020; Apolo-Apolo, Pérez-Ruiz, et al., 2020; Beltrán-Esteve & Reig-Martínez, 2014; Girshick, Donahue, et al., 2014; Hodgson et al., 2016; Martínez et al., 2016; Méndez et al., 2019; Rivas et al., 2018). The above studies have proved that using UAVs for agricultural tasks is an efficient and promising method. Images from UAVs can provide unique and rich perspectives and information, so as to realize the detection and counting of citrus fruits in large-scale unstructured orchards. However, in most studies, the counting results are obtained through the offline processing of UAV images or videos acquired in advance. This kind of method often has shortcomings such as a complex operation process and inability

to work in real-time. In addition, existing research has focused on the development of algorithms, while ignoring the embedding of data transmission between sensors and processors in the counting system, which makes it impossible to achieve a complete end-to-end production estimation system. These methods are not only complex but are also unable to directly make corresponding adjustments during field experiments to obtain the optimal counting accuracy.

Therefore, in the actual operation process, we found it necessary to develop an end-to-end yield estimation system for the following three reasons:

① Adjustment in field operation: Different orchards have different unstructured environment characteristics, such as differences in tree height and shape, fruit tree density, fruit shape and color, light conditions, wind speed, and other factors. In the actual counting operation, according to the characteristics of the orchard environment, we often need to make multiple adjustments (such as flight speed, camera distance, and flight height) to obtain the best estimation accuracy. However, when we make corresponding adjustments to the collected video and then move to the orchard's on-site laboratory, it is often difficult to ensure that the above environmental factors are consistent with the previous video collection. This will lead to difficulty in obtaining the best estimation accuracy in line with the on-site experimental environment. By using the end-to-end production estimation system proposed in this study, we can not only greatly reduce the processing time required to obtain accurate results, but also directly and quickly make corresponding adjustments on site to obtain the optimal production estimation accuracy (Apolo Apolo, Martínez-Guanter, et al., 2020).

② Timeliness of counting operation: The time from citrus maturity to picking is generally 30–50 days. If the fruit is harvested too late, it can become soft and vulnerable to damage, and there may be increased fruit dropping due to the decline of the peel hardness. In addition, late harvesting affects the storage and shelf life of fruits, causing greater economic losses (Gao et al., 2022; INRA & DIJON, 2015). Therefore, there is a requirement for us to complete high-precision orchard yield estimation more quickly, so as not to delay citrus picking, sales, and the formulation of related strategies. It should be noted that the time required for counting includes not only the time for algorithm processing but also the time for hardware adjustment and algorithm optimization mentioned in point ①.

③ Later extension: The realization of an orchard counting system with multiple UAVs can not only greatly improve the operation efficiency of yield estimation, but can also help to integrate the UAVs into a complete closed-loop platform and constantly improve the system accuracy. In addition, we can combine the system with other intelligent agricultural systems, such as fruit-picking robots, orchard intelligent navigation cars, and so on, to feedback effective information in real-time so as to better realize intelligent operations in orchards.

Unlike previous studies, in this study emperor citrus and orange were taken as the experimental objects, and an online detection and counting pipeline of unstructured orchard citrus fruits based on a UAV was proposed. Several key components of the system were

proposed and combined into an end-to-end fruit-counting system. In addition, relevant experimental evaluation was performed and the proposed method was compared with other advanced methods. The technical contributions of this study are as follows.

- (1) As far as we know, this research is the first to realize an end-to-end orchard online counting pipeline based on a UAV, which can quickly realize the development, deployment, and application of a large-scale citrus orchard counting system. This can greatly reduce the processing time required to obtain accurate results and can help farmers to make timely onsite adjustments to obtain accurate results.
- (2) Based on YOLOV5, a detection network named Citrus-YOLO was proposed, and it can effectively detect overlap, occlusion, and small-sized fruit.
- (3) A nonuniform distributed counter (NUDC) was proposed, and it can effectively suppress repeat counts caused by tracking failure and UAV flight instability.

2 | RELATED WORK

Large-scale fruit counting in orchards is an important but challenging task due to severe occlusion, perspective distortion, complex lighting, and different distributions of target sizes. In recent years, many works have been proposed to solve fruit counting problems in various situations (Chen et al., 2020). The counting methods of these works can be divided into those based on density maps and those based on object numbers.

- (1) *Counting based on density maps*: Some scholars developed methods based on deep learning to estimate the number of humans or pigs in a single image density map (Sindagi & Patel, 2017; Tian, Guo, et al., 2019; Zan et al., 2018). These methods expressed the counting problem as a density map estimation, which has the advantages of dealing with server occlusion and perspective distortion. Methods based on density maps will lose detailed personal information and discard the accurate location information of each target, resulting in losing the ability to correlate targets across time and being unsuitable for video-based counting.
- (2) *Counting based on object number*: With the development of computers, cameras, and image analysis technology, many scholars have developed fruit-number-based methods to estimate crop yield (Apolo-Apolo, Pérez-Ruiz, et al., 2020). Accurate detection and counting of fruits are the keys to realizing intelligent yield estimation in orchards. To achieve this goal, the following two difficult problems need to be solved.
 - (a) *Fruit detection*: Most of the current research has achieved the detection of various fruits using traditional image processing techniques, machine learning, or deep learning methods. However, for orchard yield estimation operations, it has been found

that in unstructured orchard environments, there are complex lighting conditions such as direct light and backlight, and diverse fruit growth states such as fruit overlapping and blocking by branches and leaves, resulting in great difficulties for the accurate detection of fruits (Zheng et al., 2021). Although deep learning algorithms have good robustness and generalization ability, the accuracy of existing fruit detection algorithms needs to be further improved because of the above difficulties and the important prerequisite of accurate orchard yield estimation, that is, accurate fruit identification.

- (b) *Fruit counting*: Most of the fruit counting algorithms proposed in existing studies perform fruit counting using a single image and cannot be applied to fruit counting for a single tree or a whole orchard. In recent years, some scholars have photographed fruit trees from multiple visual angles, and then used algorithms to detect and count fruits offline. However, the accuracy of such methods often failed to meet the requirements, due to the problem of repeat counting of fruits. In addition, many scholars have tried to use three-dimensional reconstruction algorithms, such as SFM and SLAM, combined with machine learning or neural network technology to achieve fruit counting in orchards. However, the equipment used in such methods was expensive, and the real-time performance of the algorithms was difficult to guarantee. Therefore, a new fruit-counting method needs to be proposed to ensure the accuracy and real-time performance of the algorithm, which is very important for deployment and application in practical work.

2.1 | Fruit detection

In precision agriculture, machine learning is one of the most widely used decision-making technologies in yield estimation, soil management, plant disease management, and so on. In the past decade, machine learning has been utilized in fruit yield estimation in a number of studies. Payne et al. (2013) proposed a pixel segmentation method for mangoes based on RGB and YCbCr, and conducted texture segmentation for adjacent pixels. The experimental results showed that the squared correlation coefficient (r^2) values calculated for four different angle images and a single angle image were 0.91 and 0.74, respectively. Dorj et al. (2017) used a watershed algorithm to segment citrus fruits after converting RGB images into HSV color space, achieving an r^2 value of 0.93. Lu et al. (2018) proposed a method to detect immature citrus fruits in tree images using texture and intensity distribution, with an algorithm accuracy of 80.4%. Wang et al. (2018) developed a recognition and calculation method for immature green citrus fruits, combining the LBP features and an Adaboost-based classifier, with an accuracy of 85.6%. He, Xiong, et al. (2020) proposed a green citrus fruit detection method based on a deep bounding box regression forest (DBBRF), with an average identification accuracy and a time of 87.6% and 0.759 s, respectively.

The above studies mainly combine machine learning and basic features such as color, texture, and shape to realize fruit detection

and calculation. However, in an actual complex orchard environment, the accuracy of fruit detection may be reduced by using these methods due to the influence of natural light, leaf occlusion, fruit overlap, and other factors (Lin et al., 2019; Rosebrock, 2016). Therefore, the robustness, applicability, and real-time performance of machine learning algorithms are poor in practical applications.

In recent years, deep learning has replaced traditional computer vision technology as the latest technology for fruit detection, segmentation, and counting with single images. Lin et al. (2019) identified green citrus fruits based on color and thermal information and the Faster RCNN algorithm, with a recognition accuracy of 95.5%. Zheng et al. (2018) proposed a real-time vegetable recognition system based on a deep-learning network. The system effectively identified seven vegetables and handled real-time detection tasks. Tian, Yang, et al. (2019) used DenseNet to deal with a low-resolution feature layer of the YOLOV3 network, with feature propagation being effectively enhanced and network performance being improved. Lu et al. (2019) proposed a citrus recognition method based on an improved YOLOV3-LITE lightweight neural network, introducing a GloU border regression loss function and using MobileNet-V2 as the backbone network, with an average accuracy of 91.13%. Xiong et al. (2020) proposed a visual detection method using a UAV and the YOLOV2 model to rapidly detect green mangoes on the canopy surface, with the results showing that the error rate was 1.10% by comparison with the fruit numbers of 10 mango trees calculated using manual counting and model prediction. Liang et al. (2020) first used the YOLOV3 model combined with the UNet network to segment fruit stem pixels, with an average detection accuracy of 96% for litchi and a segmentation accuracy of 95% for fruit stems. Yang et al. (2020) proposed an integrated system for detecting citrus fruits and branches and estimating their sizes; Mask RCNN was used as the citrus fruit recognition model with an average recognition accuracy of 88.15%.

The above studies show that the research on fruit detection has gradually transitioned to deep-learning-based methods due to their powerful feature extraction capabilities. However, so far, research on the application of deep learning methods to UAV videos in orchard scenes is still in the early stages. In addition, the recognition accuracy and real-time performance of the existing methods still need to be further improved.

Unlike the above methods, a network named Citrus-YOLO was proposed in this study to solve the detection problems in respect of small, overlapping, and occluded citrus fruits from the viewpoint of a UAV, and TensorRT was used to compress the model size and speed up the algorithm for easier deployment.

2.2 | Fruit counting

In recent years, many scholars have carried out research in respect of fruit counting for a tree or an orchard. Wang et al. (2019) used a YOLO network to detect mangoes and a Kalman filter to track mangoes, achieving real-time counting of mangoes with a recall rate of 62%. Vasconez et al. (2020) used Faster R-CNN and SSD to detect

a variety of fruits and used a Kalman filter to trace and remove duplicate fruits. The method could only be used to calculate dozens of fruits with low accuracy. Based on the Euclidean distance and intersection ratio, Gao et al. (2021) used the improved Hungarian algorithm and YOLOV4-tiny to perform apple matching. The experimental results showed that the tracking confidence was greatly enhanced after algorithm improvement. However, this method did not work for a long time and only performed counting for videos within 30 s. Mekhalfi et al. (2020) proposed a counting method using LAB color space and Adaboost to determine the location of kiwi fruit. The disadvantages of this method were its inability to achieve real-time operation and its poor robustness.

Some scholars have attempted to combine three-dimensional spatial information to achieve the task of fruit counting. Apolo-Apolo, Martínez-Guanter, et al. (2020) and Apolo-Apolo, Pérez-Ruiz, et al. (2020) developed a fruit detection system based on UAV image reconstruction, with a standard error of 6.59% for fruit counting. Although this method achieves good counting results, many expensive commercial software and high-performance computers are used to process precaptured UAV images. However, the operation of the method was complicated and it was unable to meet the requirements of real-time working. Häni et al. (2020) first used the Gaussian mixed model (GMM) algorithm to cluster and segment apples, then used the SFM to reconstruct the point cloud of fruits and finally completed fruit counting for an orchard. Liu et al. (2018) developed a fruit counting pipeline using a monocular camera to achieve fruit tracking and eliminate repeating fruits with FCM, a Kalman filter and the SFM algorithm. Gené-Mola et al. (2020) achieved accurate counting of fruits on both sides of a row of trees utilizing mask R-CNN and SFM. Méndez et al. (2019) used 3D lidar for modeling and the K-means algorithm for fruit detection and counting. In the case of tree pruning, there was a significant regression relationship ($R^2 = 0.63$, $p = 0.01$) between the actual number and the number determined by the algorithm. However, the algorithm was slow and unstable in a complex environment. Although the above methods can effectively eliminate duplicate counting of fruits using 3D information, they had the disadvantages of being slow, unable to operate in real-time and requiring expensive sensors.

In addition, some scholars have proved that images captured by UAVs can be used to monitor and count animals and fruits (Apolo-Apolo, Martínez-Guanter, et al., 2020; Apolo-Apolo, Pérez-Ruiz, et al., 2020; Hodgson et al., 2016; Méndez et al., 2019; Rivas et al., 2018). The counting of orchard fruits based on aerial views is a promising approach. Counting of fruits in large-scale unstructured orchards is difficult without developing algorithms for scenarios such as complex lighting, small targets, severe occlusion, overlapping, and double-counting trajectories.

In contrast to these works, we learned from the approaches proposed by Liu et al. (2018), Wang et al. (2019), and Vasconez et al. (2020), whose systems detected and calculated the number of apples in a single image and tracked fruits with image series using estimated camera movement. Based on these methods, we proposed a new citrus counting system based on a UAV with abilities of accurate online detection and counting for a large-scale unstructured citrus

orchard. In this work, an Android SDK application was first developed and a Nginx live server was built to transmit online UAV images. Then, the counting pipeline was developed to achieve accurate detection and fruit tracking using the proposed Citrus-YOLO network and association algorithm, and a counting scheme called the NUDC was proposed to obtain an accurate fruit count. Finally, to improve the efficiency of the system, we achieved the simultaneous operation of multiple UAVs. This is the first approach for the online counting of orchard fruits using UAVs; it can provide technical support for fruit counting and intelligent yield estimation in respect of large-scale unstructured orchards.

3 | MATERIALS AND METHODS

A schematic diagram of the proposed citrus counting system for large-scale orchards is shown in Figure 1. The pipeline consists of a UAV system, a control transmission system, and a visual counting system. During the counting operation, the UAV system transmits high-quality video to Android devices LightBridge, OcuSync, and other transmission protocols. Then, the control transmission system pushes the video stream to the Nginx server. Finally, the visual counting system processes the video stream to realize the detection and counting of orchard fruits. The details of each process will be introduced later.

3.1 | Hardware

The UAV models used in this study were the Phantom 4 Pro v2 and the Phantom 4 Advanced (DJI Technology Co., Ltd.). The Phantom

4 series comprises popular and intelligent consumer-grade aerial photography UAVs. DJI GO 4 software on a mobile phone can not only be used to obtain flight and sensor parameters, and so on, but it can also be used to customize the RTMP interface to push streams to third parties for live broadcast services. The idea of live broadcasting was used to complete the online collection and transmission of citrus videos. A personal PC with an NVIDIA GeForce RTX 3060, 11.0 GRAM and Ubuntu18.04OS was used as the image acquisition platform.

3.2 | Data preparation

In this study, video datasets of two orchards were constructed. To obtain an extensive and clear video of the fruit trees, the flying height of the UAV and the distance between the UAV and the fruit trees were set according to the restrictions of the orchards. The citrus orchards are located in Zengcheng District, Guangzhou City, Guangdong Province, China (23°16'N, 113°51'E). The video data collection method for the citrus orchards is shown in Figure 2a. The citrus fruits were numerous and small, with a yellow color; the variety was Emperor Citrus. The video collection experiment was carried out in October 2021, and a total of 30 UAV videos were collected. The orange orchard is located in Baiyun District, Guangzhou City, Guangdong Province, China (23°18'N, 113°22'E). The video data collection for orange orchard is shown in Figure 2b. There were fewer orange fruits and they were large with a yellow-green color; the variety was orange. The video collection experiment was carried out in April 2022, and a total of 20 UAV videos were collected. Each video collected from the two orchards contains 5–10 trees, including 300–600 fruits. The length of the videos varies from 2 to 5 min. We

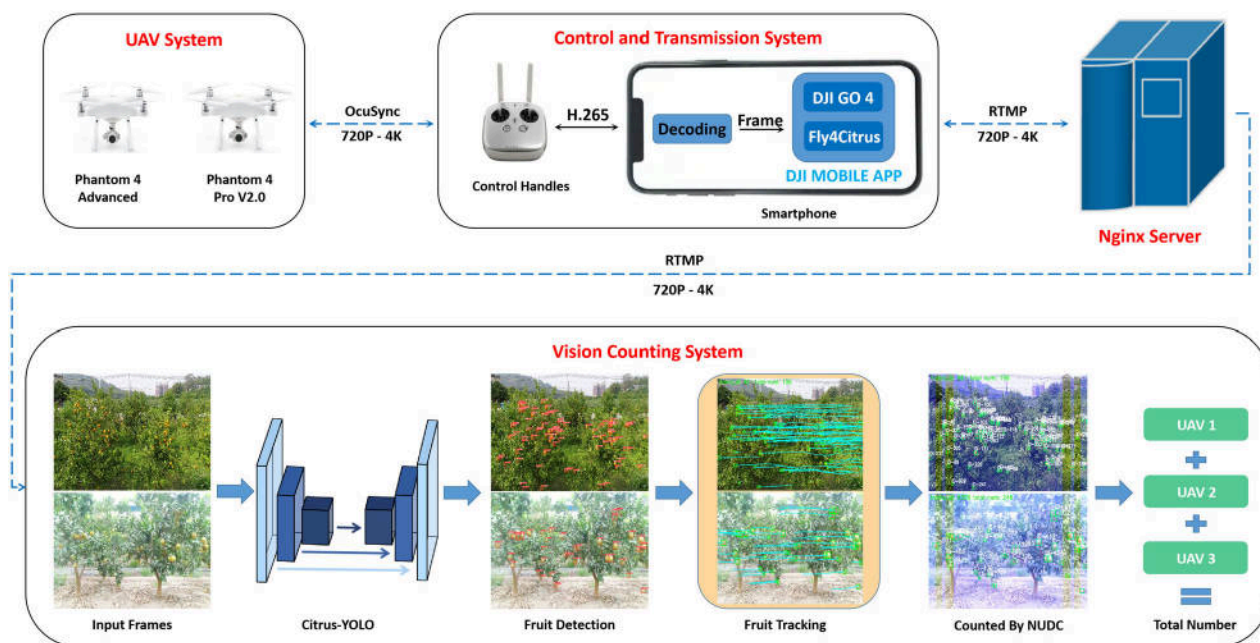


FIGURE 1 Schematic diagram of the proposed citrus counting system



FIGURE 2 Scenes of field experimentation of unmanned aerial vehicle videos acquisition. (a) Emperor citrus chard and (b) Orange orchard.

randomly divided these videos into two subsets: 35 for training and validation, and 15 for testing. Since the citrus detection network was fed with image data, the method of extracting frames from the videos was adopted and a total of 12,640 training images were obtained. Furthermore, to reduce labor and time spent on image labeling, the ground truth was obtained by labeling videos with DarkLabel.

3.3 | Vision solution

Starting from video streaming acquisition by the UAV's onboard camera, four stages were required to achieve citrus counting in the orchard. In the first stage, a UAV live broadcast platform was built to transmit high-quality online videos. In the second stage, the Citrus-YOLO network was used to detect citrus. In the third stage, citrus were tracked across frames using a Kalman filter, and the detection results were matched to trajectories using a combination of IoU, depth appearance, and Mahalanobis distance. In the fourth stage, the proposed NUDC was adopted to realize the global calculation of the number of citrus in the orchard.

3.3.1 | Stage 1: The UAV live broadcast platform

Online acquisition of UAV video is an important prerequisite for subsequent related processing. With the continuous development of information technology and communication technology, the rise of live broadcast platforms could provide a suitable solution for this research. Based on the mainstream open source architecture Nginx-rtmp-module, we built a UAV live broadcast platform for the orchard scene. The architecture of the live broadcast platform is shown in Figure 3 and includes three modules: the pushing flow end, the flow media server, and the pulling flow end. The three modules were connected in turn by the RTMP push protocol and HLS pull protocol. In this study, the DJGO 4 application developed by Shenzhen DJI Co. Ltd. was used as the pushing flow end. The Nginx server was adopted

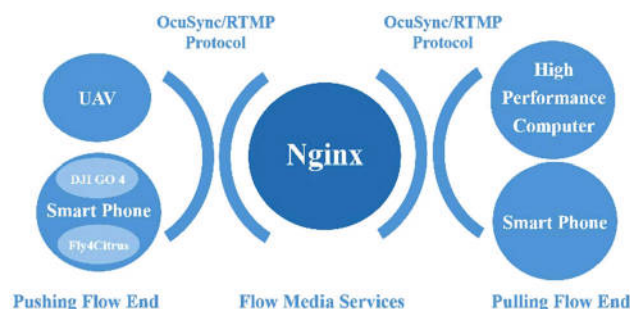


FIGURE 3 The architecture of the unmanned aerial vehicle (UAV) live broadcast platform for orchards. It includes three modules: the pushing flow end, the flow media server, and the pulling flow end. Each module is connected by the OcuSync/RTMP streaming protocol in turn.

as the flow media server. Nginx is a high-performance web server platform based on the BSD open-source protocol. It has the characteristics of less resource occupation, high stability, and large data processing capacity. A computing platform was used as the pulling flow end.

In the actual counting work, Wi-Fi was first enabled through the mobile phone to ensure that the computing platform and the mobile phone were in the same intranet. Then, through the LightBridge/OcuSync protocol, video captured by the UAV's on-board camera was transmitted to the controller. The video streaming was pushed to the Nginx server by the self-developed SDK application installed on the smartphone. Finally, the computing platform realized the online acquisition of video streaming by accessing the Nginx server.

3.3.2 | Stage 2: Detection with deep learning

In fruit counting work, the first step is to detect citrus candidates in each video frame taken by the UAV. In recent years, anchor-based object detectors (such as the RCNN series (Girshick, 2015; Girshick

et al., 2014; Ren & He et al., 2016), SSD (Liu et al., 2016), RetinaNet (Lin et al., 2017), and YOLO series (Redmon et al., 2016; Redmon & Farhadi, 2017, 2018) and anchor-free-based object detectors (such as CenterNet [Zhou & Wang et al., 2019] and YOLOX [Ge et al., 2021]) have been widely used. At present, the YOLO series has been updated to YOLOV5, released by Ultralytics. The network reached the fastest inference speed of 2 ms per image, and the mAP for the COCO 2017 validation set reached 55.4. Compared with YOLOX, YOLOV5 can quickly detect targets while ensuring that the accuracy is similar to the former. In addition, YOLOV5 has the advantage of fast fitting during training, making it more suitable for field application environments. Nonetheless, existing object detection algorithms need further improvement when carrying out UAV-based orchard counting work. From the viewpoint of the UAV, the citrus fruits in the videos not only have high density, serious occlusion and overlap, but also occupy a small number of pixels, which is challenging for YOLOV5.

As shown in Figure 4, we propose Citrus-YOLO to overcome the above limitations. The improvements included replacing PANet with BiFPN in the feature enhancement and fusion network and adding a small object detection head to the prediction network. A more efficient multiscale fusion method is adopted in the improved detection network, which can prevent the loss of small target information. The proposed Citrus-YOLO network effectively solves the shortcomings of YOLOV5 in the missed detection of occluded citrus and the false detection of background objects.

Replacing PANet with BiFPN: PANet (Liu et al., 2018) was adopted as a parameter aggregation method in the YOLOV5. Although PANet

proves the feasibility of bidirectional fusion, it ignores the fact that the contribution of features at all levels may be different, and the bidirectional fusion method used is relatively simple. Inspired by Tan et al. (2020), we improved PANet with more efficient reuse and fusing of multilayer features, as shown in Figure 5b.

Compared with PANet, BiFPN has two improvements: (1) PANet is simplified by deleting redundant nodes at both ends of the network to reduce unnecessary calculation; (2) jump connections are added between original feature nodes and output nodes at the same layer of the network, and the reuse and fusion of multilayer features are strengthened.

Increase small-object detection head: YOLOV5 performs multiple downsampling operations, which makes it difficult for deep feature maps to learn the feature information of small targets. To increase the robustness of YOLOV5 for small citrus fruits, we added a small-object detection head to the network after splicing the shallower feature map and the deep feature map.

Taking the network input of 608×608 as an example, the downsampling in YOLOV5 is used five times, so the final feature map sizes are 19×19 , 38×38 , and 76×76 . In the three feature maps, the largest (76×76) is responsible for detecting small targets. Corresponding to 608×608 , the receptive field of each feature map has a size of $608/76 = 8 \times 8$. If the width or height of the target in the original image are less than 8 pixels, it is difficult for the network to learn the feature information of the target. In addition, the resolution of images from the UAV is very large, and the downsampling factor of YOLOV5 is relatively large; it is therefore difficult for the network to

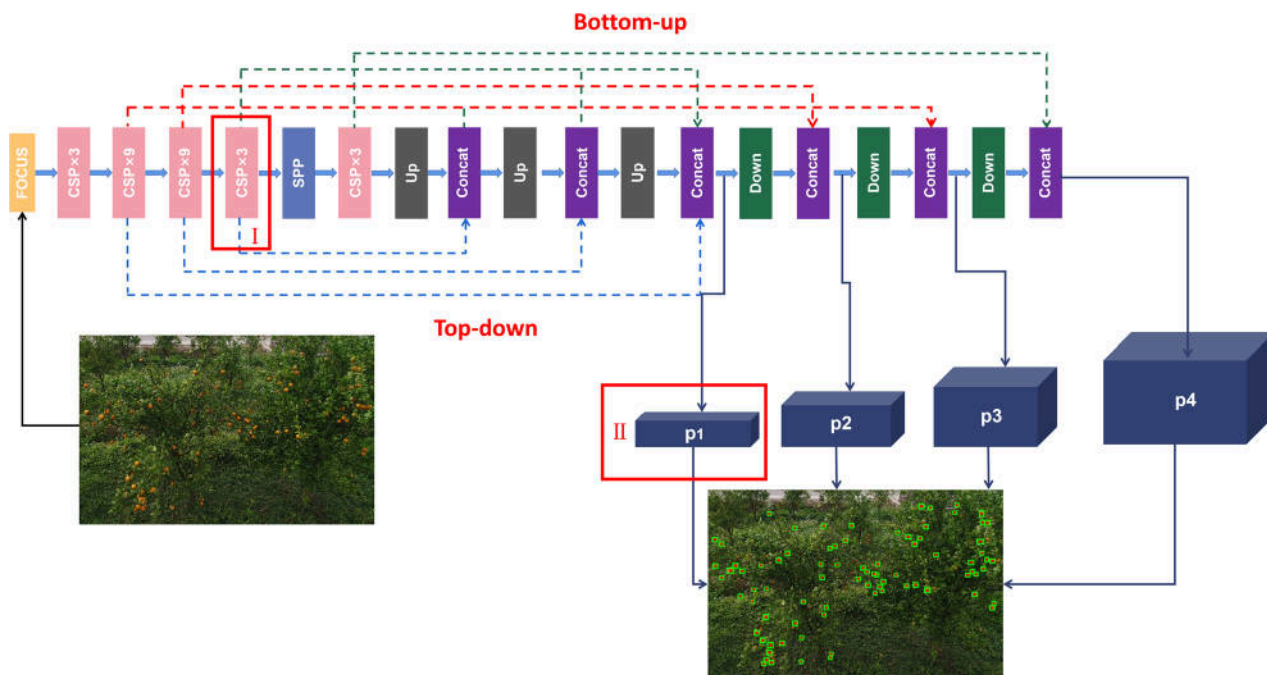


FIGURE 4 Schematic diagram of the Citrus-YOLO network. Red box I shows the addition of a $\text{CSP} \times 3$ module in the backbone for downsampling; red lines show the use of BiFPN in the feature enhancement and fusion network for more efficiently reusing and fusing multilayer features; red box II shows the addition of a detection head to the prediction network to achieve accurate detection of small targets and overlapping fruits.

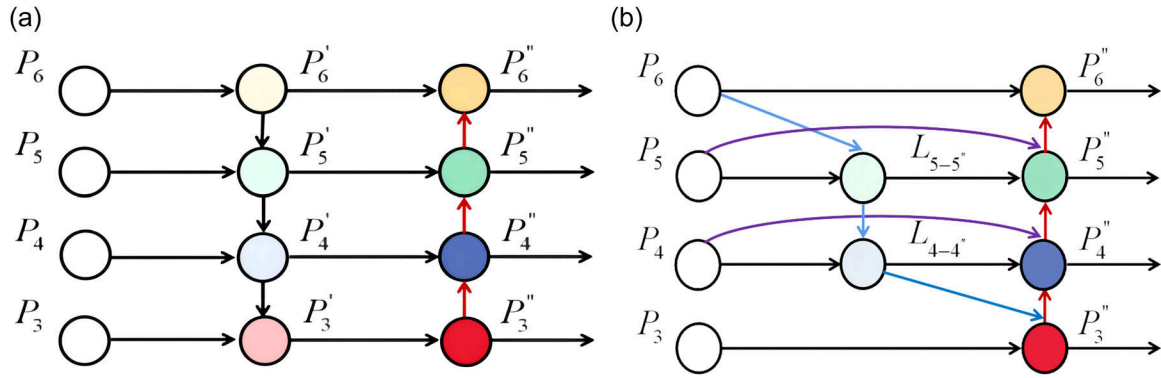


FIGURE 5 Feature enhancement and fusion network. (a) PANet and (b) BiFPN

learn the feature information of small targets for deep feature maps. Therefore, we added a small-target detection layer to shallow feature maps and deep feature maps. The feature maps were spliced and then detected to improve the detection accuracy for small targets.

3.3.3 | Stage 3: Tracking between frames

The fruit candidate region output from our deep learning fruit detection stage is used as the input for the tracking stage. This part of our pipeline is used to obtain cross-frame associations and is a necessary step to obtain fruit counts for the entire image sequence as opposed to an individual image.

After citrus fruits were detected, a Kalman filter was used to realize the cross-frame tracking of fruits. The Kalman filter is a widely used optimal tracking algorithm for linear systems. Since the frame rate of the UAV video reaches 24 frames/s, there are few changes in the position of fruit targets between consecutive video frames, and the changes in position can basically be regarded as uniform motion (Han et al., 2020). Therefore, it can be assumed that the fruit tracking system changes linearly with time. The operation of the Kalman filter includes two steps: prediction and updating of the target track (Han et al., 2020).

In the prediction stage, the Kalman filter uses estimated information for the previous video frame to predict the current frame. The results of the detection network are used to predict the corresponding fruit tracking position and its covariance matrix.

$$\hat{x}_k = F_k \hat{x}_{k-1} + Q_1, \quad (1)$$

$$P_k = F_k P_{k-1} F_k^T + Q_2, \quad (2)$$

where $\hat{x}_k$ is the prior state estimate of the k frame; $\hat{x}_{k-1}$ is the posterior state estimate of the $k-1$ frame; F_k is the state transition matrix; P_k is the prior estimated covariance of the k frame; P_{k-1} is the posterior estimated covariance of the $k-1$ frame; Q_1 represents the known external influence factor, and is set to 0 in this study. Q_2 is the covariance of the system noise, representing the influence of

external environmental uncertainty, such as wind and other factors, during the normal flight of the UAV.

In the update stage, the filter optimizes the prediction made by the filter at the $k-1$ frame through the observation value of the k frame to obtain a more accurate new estimate; that is, according to the matching matrix of the detection result and the trajectory prediction result, the fruit tracking position and its covariance matrix are updated as follows:

$$y = z - Hx', \quad (3)$$

$$S = HP'H^T + R, \quad (4)$$

$$k = P'H^T S^{-1}, \quad (5)$$

where z is the measurement value, that is, the detection result of the current frame; H is a matrix used to transform the state variable to the observation variable; Formula (3) calculates the mean error of detection and tracking, with R being the measurement noise covariance, which is denoted by a 4×4 diagonal matrix. Formula (4) first maps the covariance matrix P' to the detection space and then the noise matrix R is added. The Kalman gain K can be calculated using Formula (5) to estimate the importance of the error.

Finally, through Formula (6) and Formula (7), the mean vector x and the covariance matrix P can be updated using:

$$x = x' + Ky, \quad (6)$$

$$P = (I - kH)P', \quad (7)$$

where x' is the posterior state estimate of the k frame; P' is the posterior estimated covariance of the k frame; I is the identity matrix.

The Hungarian algorithm, a combinatorial optimization algorithm that solves assignment problems in polynomial time, was used in this study to establish the relationship between detected and predicted fruits. The matching between the fruit detections in the current frame and the predicted tracks is mainly achieved through cascade matching and IoU matching. In the cascade matching process, the prediction box and the tracking box are matched using the Hungarian

algorithm by combining motion information and appearance information.

For motion information, the Mahalanobis distance is used to describe the correlation between the Kalman filter prediction result and the detector result, shown as Formula (8).

$$d^{(1)}(i, j) = (d_j - y_i) S_i^{-1} (d_j - y_i), \quad (8)$$

where d_j and y_i represent the position of the j detection frame and the position of the i prediction frame, respectively, and S_i^{-1} represents the covariance matrix between the detection position and the tracking position. The Mahalanobis distance takes into account the uncertainty of the state estimate, where the threshold $t^{(l)}$ is set to 9.4877. If the associated Mahalanobis distance is less than $t^{(l)}$, the motion state association is successful, and the indicative function is:

$$b_{ij}^{(1)} = I[d^{(1)}(ij) \leq t^{(l)}]. \quad (9)$$

When the uncertainty of the motion information is low, the Mahalanobis distance performs well when measuring the relationship between the detection target and the track. However, when the camera view is shaken due to improper operation of the UAV by the user or due to the influence of environmental factors such as wind during the UAV flight, correlating only through the Mahalanobis distance is likely to cause the problem of citrus-ID switching, which finally leads to the repeated counting of fruit.

To solve the above problems, appearance information is added to further optimize the matching results. By using a residual convolutional neural network, the feature vector r_j of each detection target d_j is extracted, and $r_j = 1$. For tracker i , the k track is saved, corresponding to the last 100 associated feature vectors in the set $\{r_k^{(l)}\}_{k=Lk} = 1, Lk = 100$. The minimum cosine distance between the feature set of the i tracking frame and the j detection result of the current frame is calculated through Equation (10). When $d^{(2)}(i, j)$ is less than the specified threshold, the association is successful.

$$d^{(2)}(i, j) = \min \{1 - r_j^T r_k^{(l)} | r_k^{(l)} \in R_i\}. \quad (10)$$

Combining the advantages of the two measures, the linear weighting of these two measures is taken as the final measure of cascade matching. Finally, for the new fruits, as well as for detections and tracks that failed to match in the cascade matching, the IoU-based Hungarian algorithm is used to match again.

$$c_{ij} = \lambda d^{(1)}(i, j) + (1 - \lambda) d^{(2)}(i, j). \quad (11)$$

3.3.4 | Stage 4: Counting strategy

Although the large field of view of the UAV allows us to complete the counting work quickly, it also brings about changes in the citrus-ID.

When using a UAV to count citrus fruits in orchards, it is found that if the maximum value of the citrus-ID is used as the final counting result, there will be a large number of repeated counts of fruits, which will lead to great counting errors, as shown in Figure 6. To solve this problem, the single line counter (SLC), a method recently used in pedestrian detection, was introduced. However, in the actual counting work, we found that although this method can avoid the repeated counting of fruits, it has serious missed counting issues, which leads to great counting errors, as shown in Figure 7.

To solve these problems, we proposed a counting scheme named the NUDC. As shown in Figure 8, we defined dense count lines in the middle region, and the number of count lines gradually decreased from the middle to both sides. The reason for this is that the fruits in

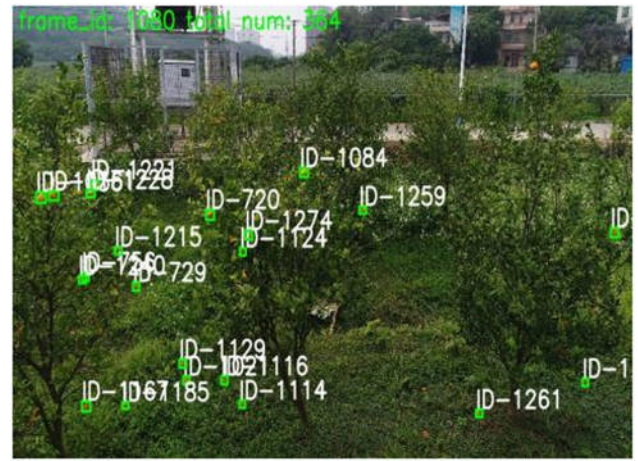


FIGURE 6 Counting based on maximum citrus-ID. The maximum ID of citrus fruits obtained by the algorithm is 1274, while the ground truth is 318.

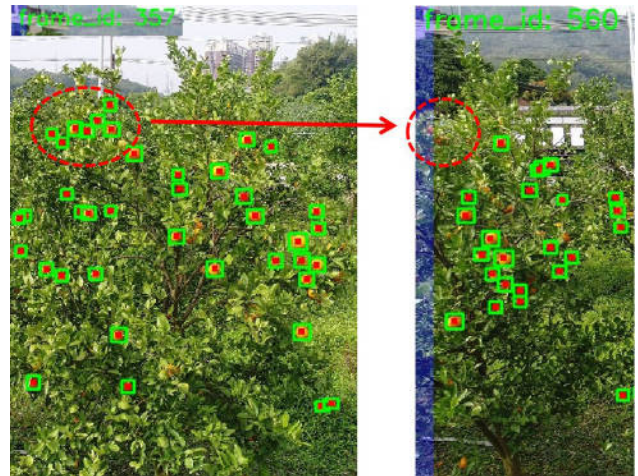


FIGURE 7 Counting based on single line counter (SLC). SLC is denoted by the blue bar in the image. It can be seen that the citrus fruits in the red circle area were successfully detected in frame 357, but failed to be detected in frame 560, resulting in a missed count.

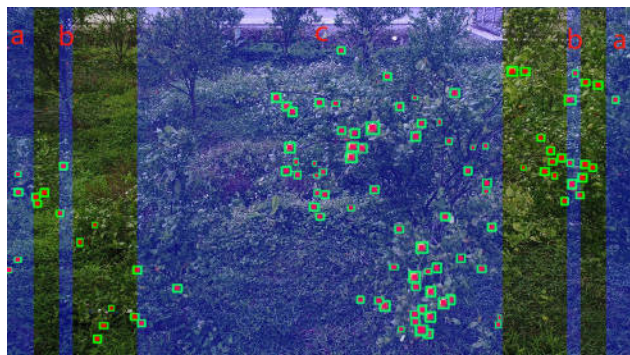


FIGURE 8 Counting based on nonuniform distributed counter (NUDC). NUDC consists of two instances of counter_a, two instances of counter_b and one instance of counter_c. Counter_a is used to count citrus fruits at the image boundary; counter_b, with a small working range, is placed on both sides of the image to count clear citrus fruits and reduce the counting of citrus fruits that fail to be tracked; counter_c is used to count most citrus fruits.

the middle of video, which are often clear and occupy a large proportion of the image, can easily be identified and tracked in a stable situation. The fruits on both sides of the image are prone to the phenomenon of citrus-ID changes, which in turn leads to repeated counting of fruits, because they are relatively far from the UAV camera and the corresponding fruits' pixels are often blurred and occupy a low proportion of image pixels. Adopting the proposed NUDC method, accurate counting results can be obtained.

4 | EXPERIMENTS AND RESULTS

4.1 | Training procedure

To properly compare the performance of the algorithms, the same training platform and similar hyperparameter settings were used for each algorithm. The running environment included a desktop computer with an Intel i7-10700 (2.90 GHz × 16) octa-core CPU, NVIDIA GeForce GTX 3090 graphical processing unit (GPU) and Ubuntu 16.04. The software included CUDA 10.0.130, CUDNN 7.4.2, NVIDIA driver 410.78, Opencv 3.4.5, Ubuntu16.04LTS, and Pytorch. Each model was trained for 100 epochs.

- (1) *YOLOV5 and Citrus-YOLO*: YOLOV5_l6.pt was adopted as the pretraining model. The size of each image was 1280 × 1280, the batch size was 32, the initial learning rate was 0.01, the momentum was 0.937 and the weight decay coefficient was 0.0005. K-means was used during training to obtain the best anchor aspect ratio. Image enhancement methods such as mosaic and flip were adopted.
- (2) *YOLOX and ByteTrack*: YOLOX_x.pth was adopted as the pretraining model. The size of each image was 1280 × 1280, the batch size was 32, the lr_per_img was 7.8125e-05, the

momentum was 0.9, the weight decay coefficient was 0.0005 and the NMS threshold was 0.65. Image enhancement methods such as mosaic, mixup and flip were adopted.

4.2 | Detection results

The performance of the detection network largely determines the effect of fruit tracking and counting. A set of comparative experiments was designed to verify the effectiveness of the proposed detection algorithm. The test comparison algorithms include YOLOX, YOLOV5, and Citrus-YOLO. Precision, recall, F1, and FPS were used as evaluation indicators. The calculation formulas are as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (13)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (14)$$

where TP represents the number of correctly detected fruits (true cases), FP represents the number of falsely detected fruits (false positives) and FN represents the number of missed fruits (false negatives). P and R denote precision and recall, respectively. F1 is the harmonic mean of precision and recall that reflects model performance. It should be noted that in this study, unlike the previous single-frame detection task, we aimed to count the fruits in the whole orchard. Therefore, the values of the above detection algorithm indicators were calculated for the whole video stream after processing, rather than simply summing and averaging the results of each frame.

Figures 9 and 10 show the detection performances of the three methods for the two orchards. The missed and false detections are manually marked with blue and yellow ellipses, respectively. Under the UAV view, our method was able to overcome the problem of missing small or occluded citrus fruits experienced by the other two networks. The UAV videos of two orchards, covering a total of 88 fruit trees, were tested with specific quantitative results, as shown in Table 1. The precision, recall and F1 values for Citrus-YOLO were higher than those for YOLO v5 and YOLOX, with respectively higher percentage points of 1.62 and 7.88 for the mean F1. In terms of real-time performance, the YOLOX ran significantly slower than YOLOV5 and Citrus-YOLO. The experimental results show that the Citrus-YOLO network has a better detection performance.

4.2.1 | Timing

The second aspect driving the choice of algorithm is its time complexity (Häni et al., 2020). As shown in Table 1, the FPS of Citrus-YOLO was 14.04 points lower than that of YOLOV5 due to

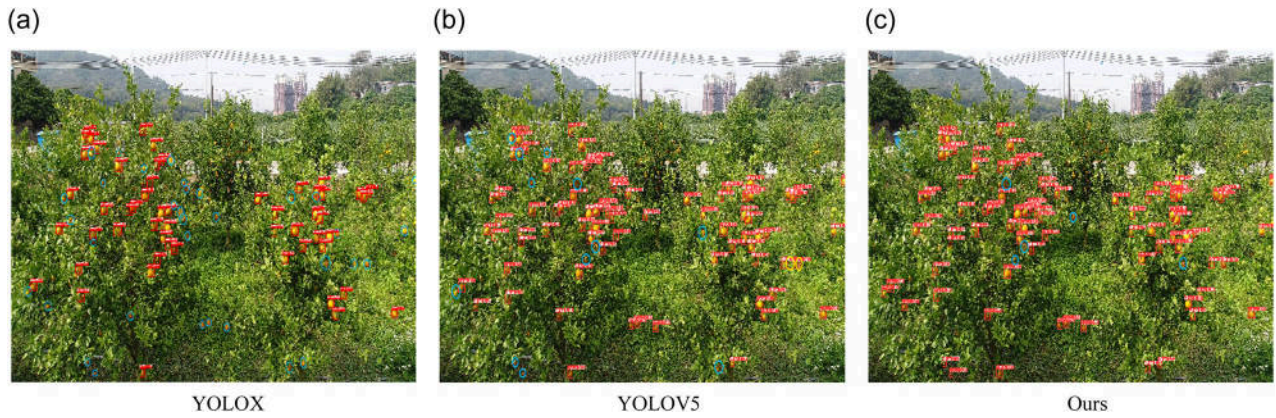


FIGURE 9 Detection results for the citrus orchard in respect of three algorithms. The missed and false detections are manually marked with blue and yellow ellipses, respectively. (a) YOLOX, (b) YOLOV5 and (c) ours.

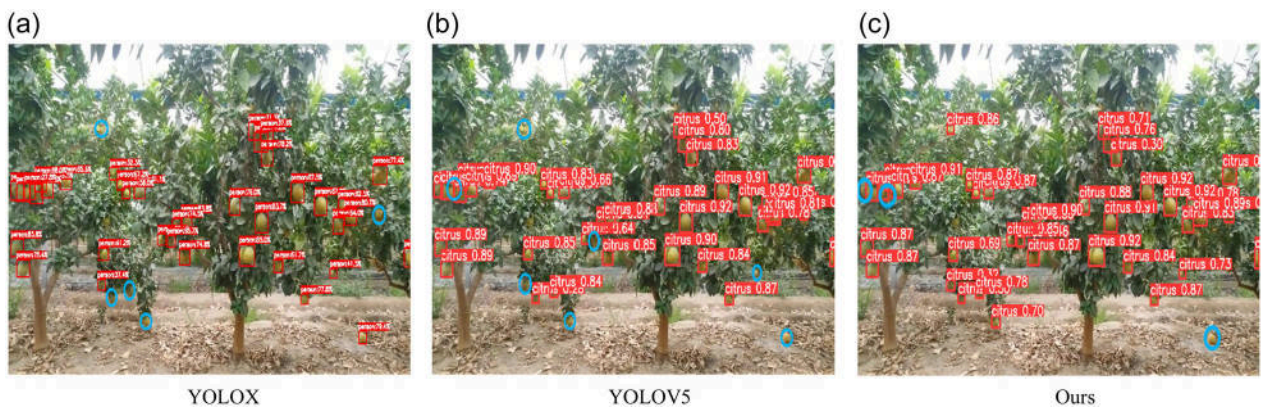


FIGURE 10 Detection results for the orange orchard in respect of three algorithms. The missed and false detections are manually marked with blue and yellow ellipses, respectively. (a) YOLOX, (b) YOLOV5 and (c) ours.

the increase in computation. To further improve the real-time performance of the detection algorithm in the inference stage, we resized the original images to 640×640 and used TensorRT to optimize the model.

TensorRT is a GPU inference engine launched by NVIDIA and only works in the inference stage. In actual counting work, resources are usually sufficient since the training is performed offline. Even if there is a temporary shortage of resources, this can be solved by increasing the training time. In contrast, the deployment of algorithms is usually more resource-sensitive. For example, computing power and memory should be considered. TensorRT is a high-performance deep-learning inference optimizer. By combining TensorRT and NVIDIA GPUs, algorithms can be deployed quickly and efficiently in most deep learning frameworks.

Table 2 shows the results for the time performance of the detection algorithm with different acceleration processes. When detecting UAV images with sizes of 4096×2160 and 640×640 pixels, the Fps values of the algorithm were 16.67 and 52.63, respectively, and the mAP values reached 96.64% and 95.25%, respectively. Compared to the detection of 4096×2160 images without acceleration, the accelerated algorithm for 640×640 images

with TensorRT (float 32) ran about nine times faster with a slight loss of accuracy. The Fps and mAP values of the model with 16-bit floating-point precision reached 144.31% and 94.19%, respectively. Compared with the method of float 16, the mAP of the model with 32-bit floating-point precision increased to 94.78%, while the Fps decreased by 1.45. In this study, method ④ was used for acceleration of the detection algorithm of the orchard counting system.

4.3 | Counting results

To verify the counting performance of our method, we compared the proposed pipeline with ByteTrack (Zhang et al., 2021) and YOLOV5 (Ultralytics, 2021). The same training, validation and testing data sets were used in the comparative experiments. Since ByteTrack and YOLOV5 did not have the counting function, we modified both of them by adding the NUDC proposed in this study. IDSR, count/ground truth, L1 loss and error mean were used as evaluation indicators, and the related calculation formulas are as follows:

$$L1 \text{ loss} = \text{Count} - \text{Ground truth}, \quad (15)$$

TABLE 1 Comparison of detection results for different algorithms

Video sequence ID	P				R				F1				FPS			
	YOLOX (%)	YOLOV5 (%)	Citrus- YOLO (%)		YOLOX (%)	YOLOV5 (%)	Citrus- YOLO (%)		YOLOX (%)	YOLOV5 (%)	Citrus- YOLO (%)		YOLOX	YOLOV5	Citrus- YOLO	
Orange orchards	95.83	94.18	94.56		84.23	92.95	94.81		89.66	93.56	94.68		8.33	66.67	52.63	
	97.16	94.31	96.29		71.40	93.09	94.59		82.31	93.70	95.43					
	93.98	91.60	91.50		68.42	94.50	94.81		79.17	93.03	93.13					
	94.87	90.68	92.45		77.89	92.34	93.01		85.55	91.50	92.73					
	97.44	95.37	95.83		93.70	88.45	90.20		95.53	91.78	92.93					
	93.71	97.28	97.38		59.03	93.35	94.24		72.43	95.27	95.78					
	95.52	96.48	96.94		73.28	91.90	93.32		82.94	94.13	95.10					
	94.90	97.17	97.36		68.89	92.84	94.06		79.83	94.96	95.68					
	97.13	96.89	96.15		84.08	94.68	96.39		90.13	95.77	96.27					
	95.57	96.51	97.54		71.32	93.75	95.82		81.68	95.11	96.67					
Orange orchards	92.13	96.38	99.43		99.15	95.00	98.31		95.51	95.68	98.87					
	95.64	98.23	98.58		96.01	92.52	95.29		95.83	95.29	96.91					
	83.91	97.98	98.18		96.05	78.23	85.04		89.57	87.00	91.14					
	84.17	98.66	98.44		97.91	82.58	84.56		90.52	89.91	90.97					
	80.69	97.34	98.35		98.71	78.21	85.24		88.79	86.73	91.33					
Mean	92.84	95.94	96.60		82.67	90.29	92.65		86.63	92.89	94.51					

$$\text{Error mean} = \frac{\text{L1 loss}}{\text{Ground truth}}, \quad (16)$$

where Count represents the result calculated by the algorithm, and Ground Truth represents the result of manual calculation.

As shown in Table 3, the average error mean values of ByteTrack and YOLOV5_DeepSORT were 71.34% and 8.22%, respectively, while the value for our method was 3.135%. The results show that our pipeline has better performance than ByteTrack and YOLOV5_DeepSORT.

It should be noted that the above data are approximate results with some errors. The reason for this is that each UAV video has about 2500 frames, and there are many fruits in each frame; this leads significant labor and time costs when calculating the above indicators. This also proves the necessity and efficiency of using computer vision technology to carry out orchard fruit counting.

4.3.1 | Analysis of experimental results

For citrus orchards, ByteTrack had the smallest IDSR compared to the other two methods. However, as mentioned above, Bytetrack and YOLOV5_DeepSORT exhibited serious missed detections of small and overlapping fruits, which led to high L1 loss. We found that the low IDSR of Bytetrack may also be caused by serious missed detections, while our method overcame this problem well. In addition, the proposed Citrus-YOLO method exhibited good detection performance for image sequences, and can effectively achieve continuous tracking of fruits.

Figures 11 and 12 show the counting results for the three algorithms before and after the stone pillar works. For the orange orchard, Bytetrack's IDSR became the largest compared to the other two methods. The reason for the results being different from those for

the citrus orchard is the presence of stone pillars in the orange orchards. As shown in Figures 11a and 12a, the IDs of the reappearing fruits were almost all changed, which led to the double counting of the fruits. Although Bytetrack algorithm performed well with the MOT20 data set, it cannot be applied to orchard counting. This is because the movement pattern of fruits is complex relative to the camera, and due to the existence of occluded objects, such as stone pillars, passing people, and so on, the algorithm cannot work well when only using Kalman filtering. In contrast, our pipeline solved this problem by adopting the DeepSORT algorithm, with the counting results of the algorithm shown in Figures 11c and 12c. Compared with YOLOV5_DeepSORT, shown in Figures 11b and 12b, our method was able to effectively detect severely occluded fruits by adopting a more efficient feature fusion method. By analyzing the counting results for the two orchards, we can see that the proposed method has better applicability and counting performance.

4.4 | Ablation study

To evaluate the effectiveness of the proposed Citrus-YOLO and NUDC, a set of comparative experiments was designed. The detection networks for comparison included YOLOX, YOLOV5, and Citrus-YOLO, and the counting strategies for comparison included Fruit ID, SLC, and NUDC. The mean absolute percentage error (MAPE) and mean absolute error (MAE) were used as evaluation metrics and were calculated as follows:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\text{Count}_i - \text{Ground Truth}_i}{\text{Ground Truth}_i} \right|, \quad (17)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n \left| \text{Count}_i - \text{Ground Truth}_i \right|, \quad (18)$$

where n represents the number of UAV videos used for testing, and is equal to 15 in this study. Count_i represents the final calculation result of the algorithm for the i th video, and Ground Truth_i represents the total number of artificially calculated fruits for the i th video.

As shown in Table 4, we found that the proposed method had the largest MAPE and MAR values when using the maximum ID of the fruit as the counting result. The reason for this phenomenon is that Citrus-YOLO can detect more citrus fruits, such as overlapping, occluded and small fruits. This also proves that Citrus-YOLO has better detection performance. When using SLC and NUDC, the MAPE and MAE of our method were the smallest. Especially when

TABLE 2 Time-consuming comparison results for different methods

Method	Acceleration process	Fps	mAP (%)
①	4096 × 2160 without acceleration	16.67	96.64
②	640 × 640 without acceleration	52.63	95.25
③	640 × 640 with TensorRT (float 16)	160.18	94.19
④	640 × 640 with TensorRT (float 32)	158.73	94.78

TABLE 3 Comparison of counting results for different algorithms

	Orange orchards (52 trees)			Orange orchards (36 trees)		
	ByteTrack	YOLOV5_DeepSort	Ours	ByteTrack	YOLOV5_DeepSort	Ours
IDSR	95	267	200	871	133	88
Count/Ground truth	2461/4362	3958/4362	4408/4362	2441/1226	1138/1226	1162/1226
L1 loss	−1901	−404	46	1215	−88	−64
Error mean	−43.58%	−9.26%	1.05%	99.10%	7.18%	5.22%

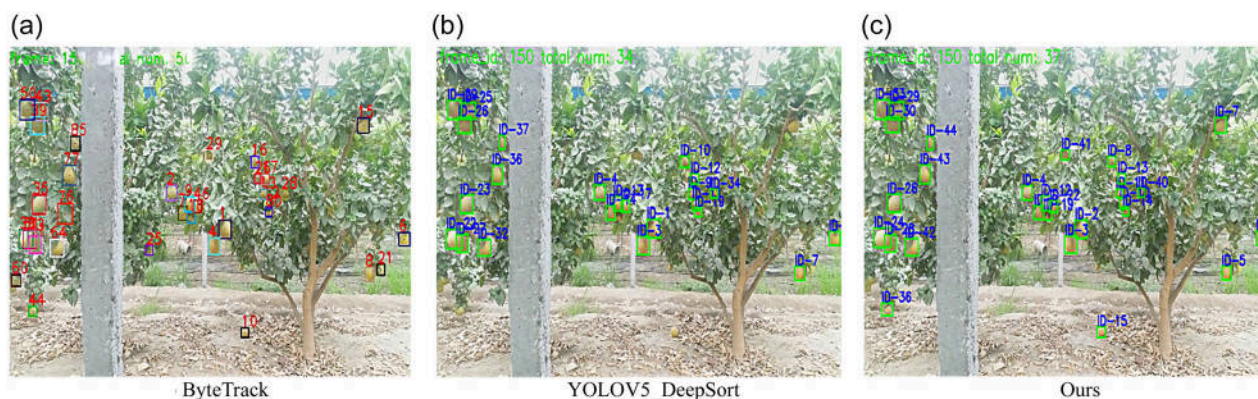


FIGURE 11 Counting results for the three algorithms before the stone pillar works. The fruits on the tree in the red box of the image are detected by the algorithm and given a unique ID. (a) ByteTrack, (b) YOLOV5_DeepSort, and (c) ours.

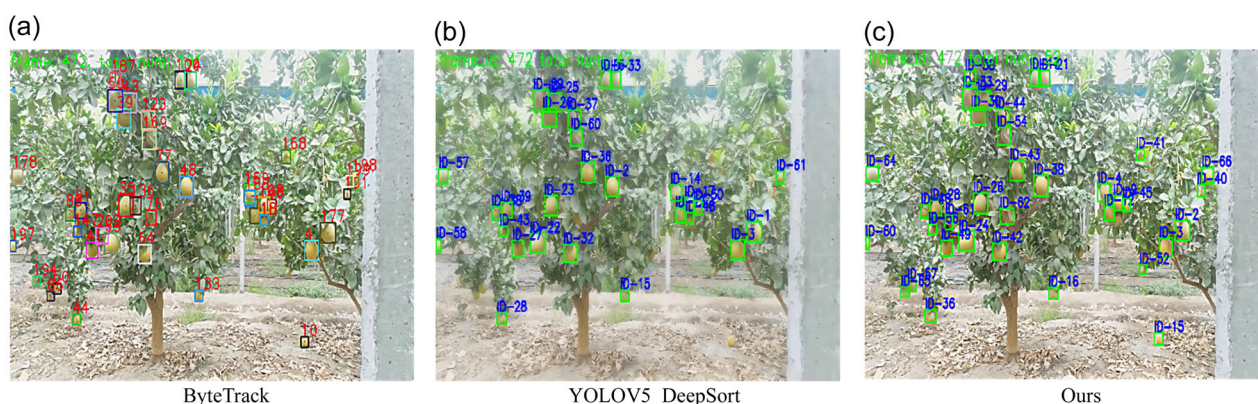


FIGURE 12 The counting results for the three algorithms after the stone pillar works. (a) The ID of most fruits on medium fruit trees has changed; that is, they are counted repeatedly. (b) Some fruits on the middle fruit tree missed inspection, leading to counting failure. (c) The algorithm stably tracks the previously detected fruits on the fruit tree, and normally detects and counts new fruits.

TABLE 4 Ablation Study

Method	MAPE (%)	MAE
YOLOX + ByteTrack	208.1	875.4
YOLOV5 + DeepSort	298.4	1375.2
Citrus-YOLO + DeepSort	383.7	1691.6
YOLOX + ByteTrack + SLC	85.3	328.1
YOLOV5 + DeepSort + SLC	75.3	306.1
Citrus-YOLO + DeepSort + SLC	70.4	281.5
YOLOX + ByteTrack + NUDC	70.1	214.3
YOLOV5 + DeepSort + NUDC	8.9	41.5
Citrus-YOLO + DeepSort + NUDC	7.1	25.9

adopting NUDC, the MAPE and MAE of our method were only 7.05% and 25.87%, respectively, which results were significantly better than those for the other two methods. The experimental results show that our method has good counting performance.

4.5 | Field experimentation

4.5.1 | DJI mobile app development

In the previous orchard experiments, we controlled the UAV using the official application DJI GO 4 installed on an Android device. This method has the following disadvantages. (1) The UAV is manually controlled, which is not conducive to the intelligent counting of orchard fruits. (2) It is difficult for operators to maintain absolute flight stability during operation, and changes in UAV speed will influence the algorithm. (3) The narrow space in the orchard and the uneven ground will to some extent influence the stable flight of the UAV. In severe cases, a UAV crash may occur, causing financial loss.

To solve the above problems, based on DJI's official Mobile SDK app (DJI Technology Co., Ltd.), we redeveloped and implemented an application named Fly4Citrus. XML language was used to design the interface of Fly4Citrus. The control function of Fly4Citrus was implemented based on Dajiang's LiveSteamView, FlightController, SetGetISOView and other classes. As shown in Figure 13, when the

UAV is working, the user first fills in the streaming address and clicks the Start Streaming button. Then, the RTMP protocol in the LiveStreamView class is called to encode the video stream into a binary stream and transmit it to the Nginx server specified by the streaming address. Audio is turned off by default to reduce bandwidth consumption. Users can press the Open/Close Sound button to switch when necessary. The Change ISO button is designed to change the sensitivity and aperture size of the UAV camera to adapt to different lighting environments, so as to obtain a video with moderate brightness. Switch Video Source is used to switch between different video sources. Through the FlightController class interface, we have achieved command control of UAV autonomous flight, replacing the previous manual control. The UAV is controlled to fly stably and photograph the orchard trees by setting the flight distance, flight height, flight speed and moving direction according to personalized

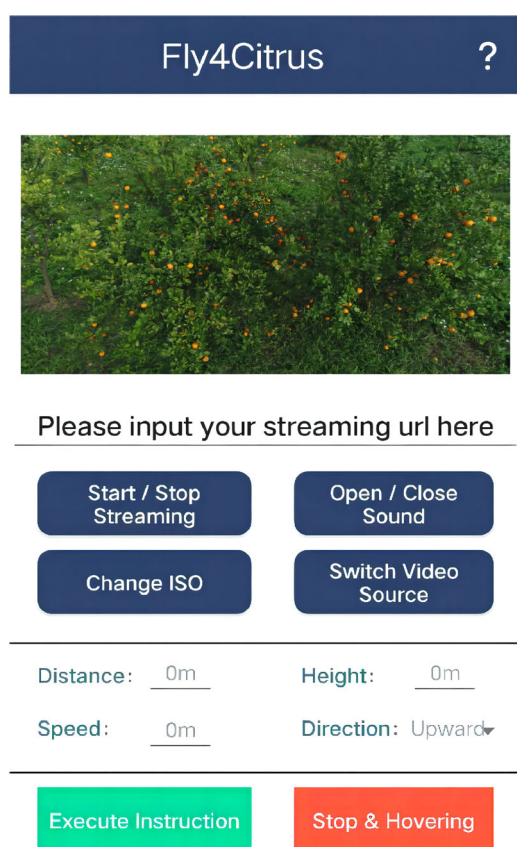


FIGURE 13 Fly4Citrus app. Stable flight of unmanned aerial vehicle and online acquisition of video stream can be realized through button switching and parameter setting in the app.

requirements. Finally, the proposed counting pipeline accesses the Nginx server to obtain the video stream online.

4.5.2 | Comparison experiment of offline and online counting

Although the UAV live broadcast platform can realize the online transmission of UAV images, the live broadcast scheme is inevitably affected by the quality of the communication network in an orchard environment. Therefore, to explore the impact of the UAV live platform on the performance of the counting system, a set of comparative experiments was designed based on offline video and online transmitted video in this study. The comparative test objects were four rows of fruit trees, comprising 33 fruit trees. Delay, packet loss rate, IDs and MAPE were used as indicators in the experiments.

As can be seen from Table 5, in terms of UAV image data transmission performance, compared with the offline counting mode, the online counting mode had 8–15 s delay and 3.4% packet loss rate. In terms of counting performance, the average IDs and MAPE of the offline counting mode were 18% and 5.28%, respectively, while the average IDs and MAPE of the online counting mode increased to 57.67% and 14.95%, respectively.

Analysis of experimental results

During the online counting experiment, a likely reason for the difference in the counting accuracy between the two methods was that there were loopholes and packet losses of image data when the communication network was not in good condition during the process of online transmission of UAV images; that is, some image data were discarded before they were transmitted to the counting system. Figure 14 shows the counting effect diagram of the algorithm in a poor-condition communication network. When the algorithm detected the k th frame image of the UAV video as shown in Figure 14a, due to the loss of image data, it received the $(k + m)$ th frame image shown in Figure 14b at the next moment, with poor image quality that was not conducive to the algorithm for fruit detection. In addition, due to the nonconformity of the hypothesis of the constant speed movement model introduced by the tracker, the actual position of fruits in the $(k + m)$ th frame image did not match the position predicted by the tracker, resulting in the failure of some fruit tracking across frames. This phenomenon also confirms our hypothesis that when the UAV is working, the communication fluctuation of the network will have a certain impact on the stable transmission of the UAV image sequence, resulting in an increase in the counting errors during the online operation.

Method	Delay	Packet loss rate	Count			
			Prediction	Total fruit	Average IDs	Average MAPE (%)
Offline	0	0	1322	1256	18	5.28
Online	8–15 s	3.4%	1444	1256	57.67	14.95

TABLE 5 Comparison of offline and online counting results

Nevertheless, as shown in Figure 14c, we can see that in the $(k+n)$ th frame image, the ID of some fruits remained unchanged, which indicates that the proposed method has certain robustness. The final experimental results show that the error of the UAV online counting system proposed in this study was 14.95%, and the counting accuracy was within an acceptable range. In future work, we will solve this problem by optimizing the communication network quality or by using higher-quality and more efficient online transmission methods.

4.5.3 | Real-time counting experiment under different lighting conditions

Since the information in respect of fruit objects in citrus images under different illumination conditions is often different, the detection effect of the corresponding algorithm will be influenced to some extent. Therefore, fruit counting experiments in respect of 20 citrus trees were carried out under three different light intensities (strong light, suitable light and weak light), and the effects of different light conditions were evaluated. The light intensity was measured by illuminance. The fruit detection rate was still represented by P, R, and F1, and the fruit counting accuracy was represented by IDs, false detection and MAPE.

The experimental results are shown in Table 6. Under normal light conditions, the algorithm achieved the best performance, with the highest F1 value of 89.07% and the lowest MAPE value of 12.75%. Under strong and weak light conditions, the counting performance of the algorithm was similar, the F1 values were 87.19% and 87.05%, respectively, and the MAPE values were 15.99% and 15.59%, respectively.

The experimental results show that environmental factors such as strong light and weak light will increase the difficulty in respect of fruit detection and counting by the algorithm to some extent.

4.5.4 | Multi-UAV real-time counting experiment

The efficiency of most existing work is mainly guided by indirect standards of computational efficiency, such as FLOP or parameters. These circuitous standards are one-sided and suboptimal, and sometimes deviate from the deployment of the real scene. In fact, the model must deal with environmental uncertainty in the deployment process, which involves hardware perception characteristics, such as GPU memory requirements.

Specifically, as mentioned in many papers, indirect metrics of computational complexity are widely used in neural network architecture design. FLOPs and parameters are approximate, but usually cannot reflect direct metrics. In practice, what we really care about is the speed or delay of the algorithm (Xia et al., 2022).

To further improve the operating efficiency of the system and make the counting system more applicable and valuable in promotion, we explored the time and resource consumption of the counting system under different UAV numbers. The experimental objects were two algorithms with and without TensorRT acceleration.

The indicators included the total time consumption, detection time consumption and tracking time consumption of the two algorithms under different numbers of UAVs. For resource

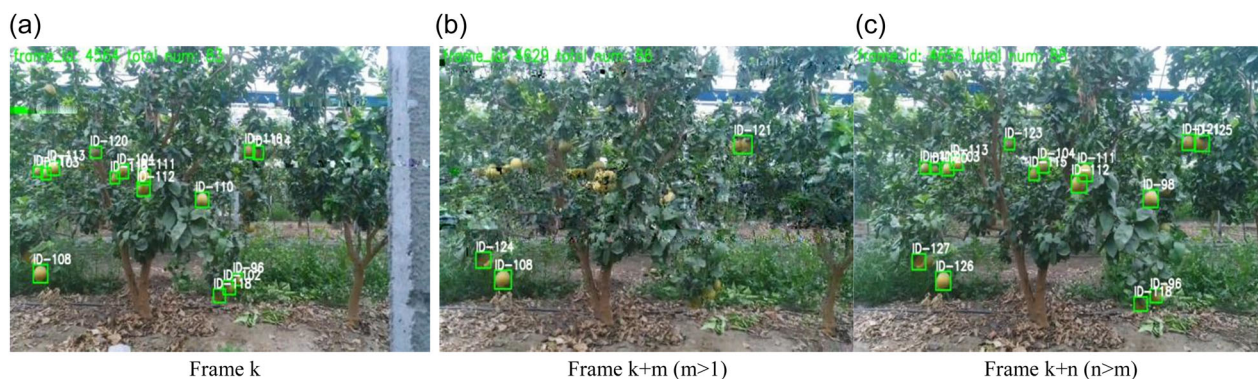


FIGURE 14 Counting results of our algorithm in the presence of poor network conditions. (a) Frame k , (b) frame $k+m$ ($m > 1$), (c) frame $k+n$ ($n > m$).

TABLE 6 Statistics of counting results under different light conditions

Light condition (kLux)	Number of trees	Ground truth	Detection			Counting			
			P (%)	R (%)	F1	IDs	False detection	Total	MAPE (%)
Strong light /(>70)	19	494	93.33	81.80	87.19%	51	26	415	15.99
Suitable light /(<25-50)			95.51	83.44	89.07%	48	18	431	12.75
Weak light /(<10)			93.80	81.21	87.05%	54	24	417	15.59
Mean			94.21	82.15	87.77	51	22.67	421	14.78

consumption, the GPU memory requirement of the two algorithms under different numbers of UAVs was analyzed.

Figure 15 shows the results for total time consumption, detection time consumption and tracking time consumption of the two algorithms under different numbers of UAVs. Comparing Figures 15a and 15d, we found that the total time consumption of the algorithm under different numbers of UAVs was greatly optimized with TensorRT acceleration. As shown in Figure 15b,e, after TensorRT acceleration, the detection time of the algorithm under different numbers of UAVs was maintained at about 0.02 s, which was 5–7 times higher than the original algorithm without acceleration. As shown in Figure 15b,e, the time consumption of the two algorithms for tracking was almost the same.

As shown in Figure 15d, the total time consumption of the counting system increases with the increasing number of UAVs. When the system worked with one, two and three UAVs, the algorithm spent about 0.05, 0.06, and 0.10 s, respectively, processing the image. The experimental results show that the proposed counting system can meet the real-time requirements of counting work when performing operations with multiple UAVs.

It can be seen from Figure 16 that the memory consumption of the counting algorithm in this study increases with the increasing number of UAVs. The apparent memory consumption values for the algorithm were 2.34, 6.50, and 8.61 g for one, two and three UAV counting operations, respectively. After using TensorRT, the memory

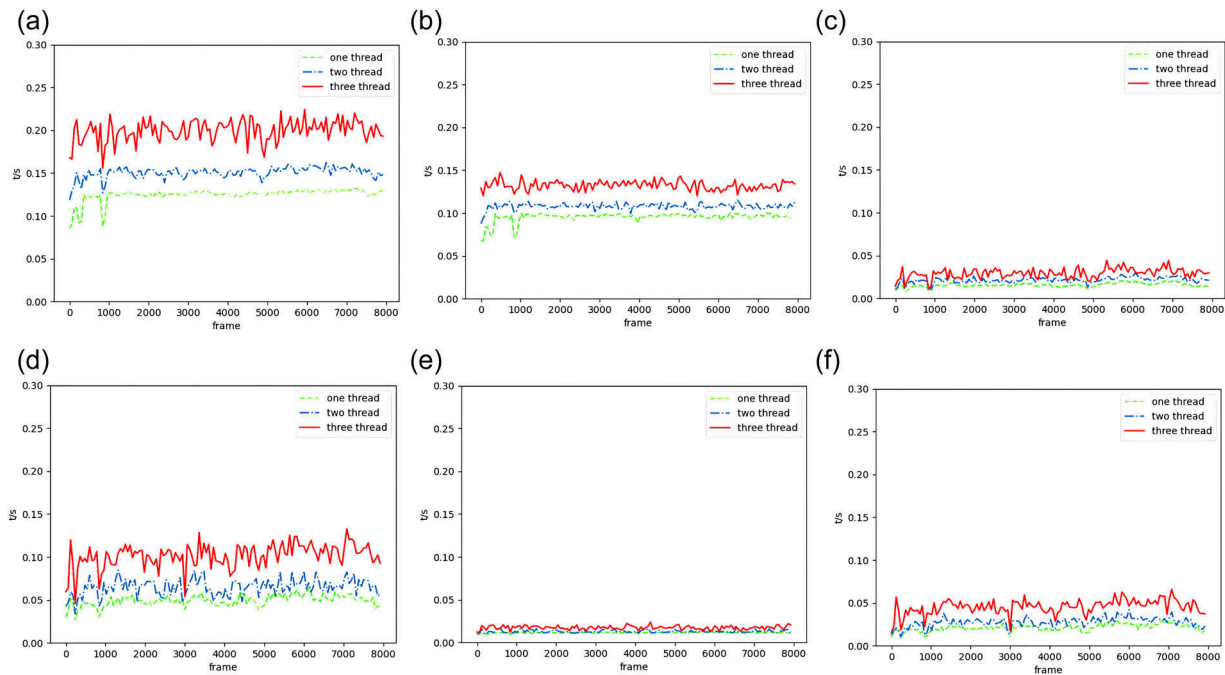


FIGURE 15 Time consumption of counting system under different numbers of unmanned aerial vehicles (UAVs). Green, blue and red respectively indicate the time consumption of the algorithm when one, two and three UAVs are working at the same time. (a) origin_all, (b) origin_detect, (c) origin_track, (d) TensorRT_all, (e) TensorRT_detect, and (f) TensorRT_track.

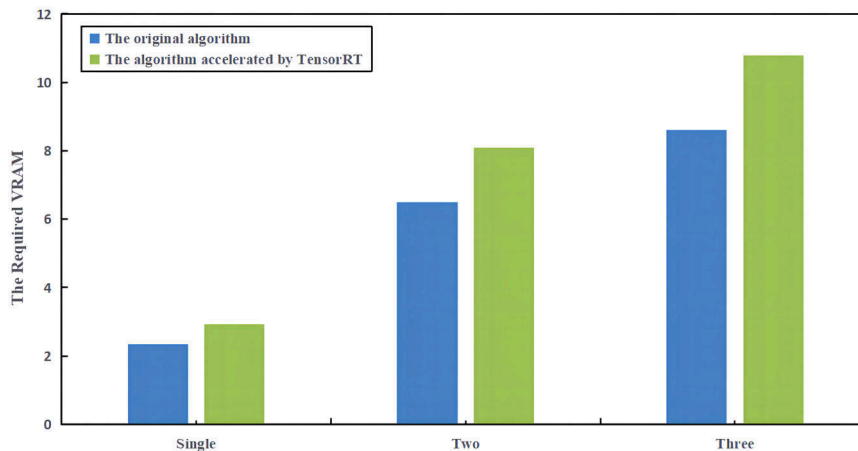


FIGURE 16 Graphical processing unit memory consumption under different numbers of unmanned aerial vehicle (UAVs). Blue and green indicate the resource consumption before and after algorithm acceleration under different numbers of UAVs.

consumptions of the algorithm were increased by 0.59, 1.59, and 2.17 g. Therefore, with sufficient hardware resources, the proposed system can realize the online counting of multiple UAVs, which can greatly shorten the time spent on fruit-counting tasks.

4.6 | Comparison with other citrus counting methods

To further verify the performance of the proposed method, we designed experiments to compare the performance of the most advanced citrus counting methods (Zhang et al., 2022) and the method proposed in this paper. The experimental comparison includes detection and overall counting. F1 value, FPS, and MAPE were used as evaluation indicators.

As shown in Table 7, UAVs were adopted by us while a field rover was adopted by Zhang et al., which means that our application scenarios have more prominent problems in detecting small fruits and overlapping occluded fruits. However, in the case of offline operation, the F1 value, MAPE and FPS of our method were 94.51%, 5.28%, and 158.73%, respectively; these values are higher than those of Zhang et al.'s method of 91.60%, 8.10%, and 83.3%. The final experimental results show that the diversity score obtained using our method is relatively better than the method proposed by Zhang et al. (2022) has better detection and counting performance.

In addition, this research addresses the shortcoming of most methods only being able to work offline. By building an orchard UAV live broadcast platform, an end-to-end orchard online counting pipeline has been realized for the first time, achieves 14.95% MAPE and 50 FPS, and has good counting and multiple UAVs can work at the same time. The experimental results show that the proposed online counting pipeline performance. At the same time, the online counting method combined with multiple UAVs working at the same time realized in this paper can not only greatly reduce the time consumption in the development, optimization and practical application of the algorithm, but also has high operating efficiency; the method can provide effective information for farmers to make decisions in a short time.

4.7 | Discussion

This study provides valuable insights into automatic counting operations for fruits in emperor citrus and orange orchards in UAV

application scenarios, especially for the construction of a UAV live broadcast platform, the detection and counting of fruits, the field operation performance and the resource consumption of the algorithm.

To address the limitation of UAVs only performing offline operations, an orchard UAV live broadcast platform based on the Nginx-RTMP-Module architecture was built to transmit UAV high-quality video streams online. Nevertheless, the UAV live broadcast platform had a delay of 8–15 s and a packet loss rate of 3.4%. When the communication network was in poor condition, the uniform motion model assumed by the Kalman filter often failed to work, which led to repeated counting of some fruits. Compared to the offline counting scheme, the online algorithm achieved 39.67% and 9.67% increases in IDs and MAPE, respectively.

The Citrus-YOLO network, using Bi-FPN to replace PAN-Net and adding a small target detection layer, was proposed for fruit detection with real-time UAV videos, and was able to effectively prevent missed detection of small and severely occluded fruits. Compared with YOLOV5 and YOLOX, Citrus-YOLO's F1 values increased by 1.62 and 7.88 percentage points, respectively. However, the Citrus-YOLO network had only 52.63 FPS due to the increase in computational work. To further improve the real-time performance of the detection algorithm in the inference stage, 640 × 640 pixels images and a model accelerated by TensorRT were adopted in this study. Compared with the model without acceleration, the FPS of the accelerated model was increased to 158.73 with a loss of 0.47% mAP.

For the counting of fruits, the DeepSORT algorithm was first adopted to achieve continuous tracking of fruits in UAV videos. Then, a NUDC was proposed to suppress repeat counting due to tracking failures or UAV flight instability, and experiments were designed to compare the counting performance of Bytetrack, YOLOV5_DeepSORT and the proposed method. The experimental objects were two orchard scenes with different characteristics. Without occluders, Bytetrack had the smallest IDSR. However, Bytetrack and YOLOV5_DeepSORT exhibited serious missed detections of small targets and overlapping fruits in the UAV field of view, resulting in their high L1 loss. In the presence of occluders such as stone pillars, compared to the other two methods, Bytetrack's IDSR was the largest. Occlusion was the essential reason for this phenomenon in ByteTrack. When fruits blocked by stone pillars reappeared, their IDs were almost all changed, leading to double counting of fruits. Although our method can overcome the above problems well, repeated counting still occurs

TABLE 7 Comparison with the state-of-the-art approaches

Counting method	Tool	Offline/online	Detection		Detection + Counting	
			F1 (%)	FPS	MAPE (%)	FPS
Zhang et al. (2022)	Field rover	Off	91.60	83.3	8.10	-
Ours	UAV	Off	94.51	158.73	5.28	50
		On	87.77		14.95	

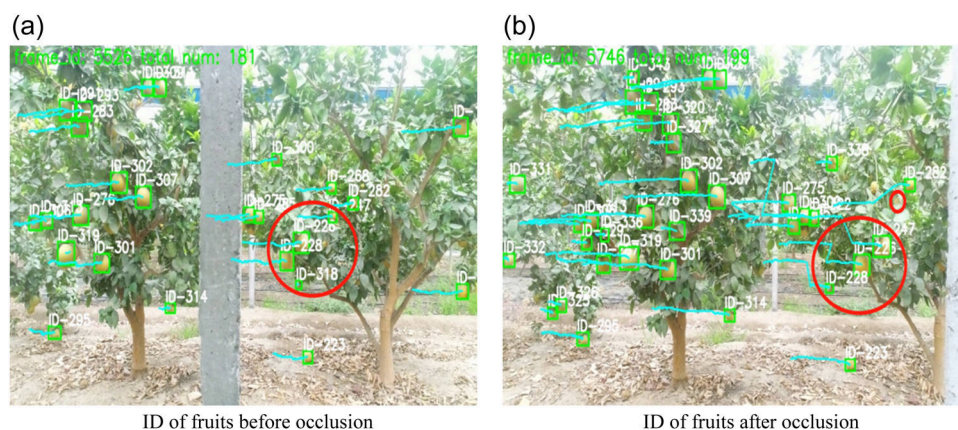


FIGURE 17 Trajectory changes between multiple fruits. The final result is not affected although the IDs of fruits in the red circle have changed. ID of fruits (a) before occlusion and (b) after occlusion.

when fruits are occluded for a long time and then reappear. The reason for this phenomenon is that the Kalman filter has instability in the prediction of fruits occluded for a long time. Interestingly, we found in our experiments that when some occluded fruits reappeared, the algorithm had a small probability of assigning them with the same IDs as before, which did not affect the counting results, as shown in Figure 17a,b. The reason for this is that the appearance features of the fruits have a certain degree of similarity, and their shooting distances are relatively short. When the fruit reappears, the Hungarian algorithm matches it to the trajectory predicted for another occluded fruit.

In addition, we only counted fruits from one side of the trees. There is no doubt that the fruits growing inside the trees can be seen from multiple angles during the movement of the UAV. However, when the branches and leaves grow densely, the algorithm cannot count the fruits severely occluded on the other side, which leads to missing detection of a small number of fruits.

The performance of the fruit counting algorithm under different lighting conditions and different numbers of UAVs was examined in the experiments. The results show that under suitable lighting conditions, the performance of the algorithm was the best, with an F1 value and MAPE of 89.07% and 12.75%, respectively. In strong light and weak light conditions, the algorithm was more prone to false detections and failed tracking, with F1 values of 87.19% and 87.05%, and MAPEs of 15.99% and 15.59%, respectively. To reflect the performance of this algorithm in actual deployment, and to strengthen the implementation and promotion value of this research, we explored time and resource consumption by the counting system under different numbers of UAVs. The total running time of the counting algorithm increases with the increasing number of UAVs. When one, two, and three UAVs were operating at the same time, the algorithm spent about 0.05, 0.06, and 0.10 s on processing one frame of UAV image, respectively. The experimental results show that the proposed counting system can still meet the real-time requirements when multiple UAVs perform counting operations simultaneously. For resource consumption, the memory required by

the proposed counting system increases with the increasing number of UAVs. The system consumed 2.34, 6.50, and 8.61 g of VRAM when the algorithm performed counting operations for one, two, and three UAVs, respectively. Especially after acceleration using TensorRT, the memory consumption of the algorithm was increased by 0.59, 1.59, and 2.17 g, respectively. Although the system proposed in this study can achieve real-time counting with multiple UAVs, the hardware resource consumption will also increase with the increasing number of UAVs.

In general, the reasons for the failure of online fruit counting are as follows: (1) When the communication network signal is poor, the orchard live broadcast platform cannot stably transmit the UAV videos, resulting in failure of fruit tracking. Optimizing the quality of the network communication or adopting a higher-quality online transmission method are problems that need to be solved in the future. (2) When the fruit reappears after being blocked for a long time, its ID often changes, which leads to repeated counting of fruit. (3) In this study, only fruits from one side of the tree were counted, and there was a problem in respect of missing detection of some fruits. Therefore, to effectively solve the problems of fruit occlusion and missing detection in the traditional single-view, research on multiview fruit online counting based on the cooperative operation of multiple UAVs will be the focus of future work.

5 | CONCLUSIONS

In this study, a new UAV fruit counting pipeline was proposed, which can accurately and quickly detect and count citrus fruits in large-scale unstructured orchards. As far as we know, this is the first orchard fruit counting method that can be realized online and support multiple UAVs working in parallel. This work can greatly reduce time consumption in the development, optimization, and practical application of the algorithm. In addition, it has high operating efficiency, and can provide technical support for the counting and yield estimation of large-scale unstructured orchards.

In the field of orchard yield estimation, existing methods can only achieve offline counting of fruits from videos. To address this limitation, an orchard live broadcast platform based on Nginx-rtmp-module architecture was built to transmit UAV high-quality video streams online. The experimental results show that the live broadcast platform has a delay of 8–15 s and a packet loss rate of 3.4%. The average IDs and MAPE of the offline counting method were 18% and 5.28%, respectively, while the average IDs and MAPE of the online counting method were 57.67% and 14.95%, respectively.

A detection network named Citrus-YOLO was designed to detect fruits in real-time from UAV video. As a result of adding one more detection head and replacing PAN-Net with Bi-FPN, the Citrus-YOLO network can effectively prevent the missing detection of small and severely occluded fruits. The experiments show that the F1 value of the Citrus-YOLO network reached a high value of 94.51% with the UAV test video of the orchard environment. Compared with YOLO v5 and YOLOX, the F1 values of the Citrus-YOLO network increased by 1.62% and 7.88%, respectively. To further improve the real-time performance of the Citrus-YOLO network, smaller images and TensorRT were adopted in this study. The experimental results show that the FPS of the accelerated Citrus-YOLO network increased to 142.86 with a loss of 0.47% mAP. The proposed Citrus-YOLO network had better detection accuracy and speed, and can effectively meet the needs of actual orchard operations.

To suppress the repeated counting caused by tracking failure or UAV flight instability, the NUDC was proposed in combination with the DeepSORT algorithm to achieve continuous tracking and global counting of fruits in UAV images. In two orchard scenes with different characteristics, an experiment was designed to compare the counting performance of ByteTrack, YOLOV5_DeepSORT and the proposed method. IDSR, count/ground truth, L1 loss and error mean were used as evaluation metrics. The experimental results show that the average error means of ByteTrack and YOLOV5_DeepSORT were 71.34% and 8.22%, respectively, while the mean for our method was 3.135%. Therefore, our method has wider applicability and better counting performance. To evaluate the effectiveness of Citrus-YOLO and NUDC, a set of comparative experiments was designed. MAPE and MAE were used as evaluation metrics. The experimental results show that the MAPE and MAE of our proposed pipeline were only 7.05% and 25.87, respectively.

Moreover, to reflect the counting performance of our method in actual deployment, field experiments in respect of fruit counting were carried out in a citrus orchard. The experimental results show that under suitable lighting conditions, the F1 value and MAPE of the algorithm were 89.07% and 12.75%, respectively. When working with one, two, and three UAVs, it took about 0.05, 0.06, and 0.10 s to process one frame of UAV image, respectively, and the required GPU memory was 2.93, 8.09, and 10.78 g, respectively.

Finally, a group of experiments was designed to compare the performance of the most advanced citrus counting methods and the method proposed in this paper. The experimental results show that the diversity score obtained using our method is relatively better than

the method proposed by Zhang et al., which shows that it has better detection and counting performance.

ACKNOWLEDGMENTS

This study was supported by the National Natural Science Foundation of China under Grant no. 32071912, the open competition program of top ten critical priorities of Agricultural Science and Technology Innovation for the 14th Five-Year Plan of Guangdong Province under Grant 2022SDZG03, and the Special Funds for the Cultivation of Guangdong College Students' Scientific and Technological Innovation under Grants no. pdjh2021a0070 and under Grant no. pdjh2022b0079.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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How to cite this article: Zheng, Z., Xiong, J., Wang, X., Li, Z., Huang, Q., Chen, H., et al. (2023) An efficient online citrus counting system for large-scale unstructured orchards based on the unmanned aerial vehicle. *Journal of Field Robotics*, 40, 552–573. <https://doi.org/10.1002/rob.22147>